Joan San Bartolomé Ferri

Digital Design Engineer (RTL) | Verilog/SystemVerilog · FPGA · UART · Embedded C



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Professional Profile

Electronics engineer completing an MSc in Electronic Systems Engineering at UPV (expected January 2027), focused on RTL design in Verilog/SystemVerilog for FPGA. Designed a real-time 48 kHz/24-bit IIR crossover filter with its own UART, verified bit-exact against a Python golden model and using 8.7 % of the FPGA's LUTs. Also built a synchronous FIFO with a constrained-random testbench (100 % functional coverage) and a Nios II SoC with custom Verilog peripherals and C firmware, and designed PCBs in Altium (4-layer) and KiCad.

Education

Master's Degree in Electronic Systems Engineering — *09/2025 – 01/2027 (expected)*
Universitat Politècnica de València — Valencia, Spain

Master's Degree in Applied AI and Production Process Optimization — *04/2026 – 10/2026*
Visual Business School — Valencia, Spain

Bachelor's Degree in Industrial Electronic and Automation Engineering — *09/2021 – 01/2026*
Universitat Politècnica de València — Valencia, Spain

Projects

Synchronous FIFO: RTL Design and Constrained-Random Verification — *09/2026 – 10/2026*
Universitat Politècnica de València (UPV) · MSc coursework
SystemVerilog (OOP), QuestaSim, Tcl, Git. Team of 2. Designed a parameterizable synchronous FIFO (WIDTH/DEPTH) in RTL from an ASM chart over dual-port RAM. Built a class-based, pre-UVM testbench with constrained-random stimulus, driver, monitor, reference-model scoreboard and covergroups, reaching 100 % functional coverage (141/141 bins) in 2,550 cycles vs 99.79 % for a fixed 10,000-cycle run, with 0 mismatches. Automated a Tcl regression (6/6 configurations) and validated the checkers by injecting 3 fault types with bind and force/release.

STM32F746ZG Control Board: 4-Layer PCB with Controlled Impedance — *02/2026 – 06/2026*
Altium Designer. Individual MSc project (grade 9.5/10). Redesigned ST's NUCLEO-F746ZG reference: hierarchical schematic, component selection, 4-layer controlled-impedance stack-up (CAN 120 Ω, Ethernet MDI 100 Ω, USB FS 90 Ω), 0–10 V ADC front-end with RC anti-aliasing filter and op-amp buffer, ESD protection, post-layout signal-integrity simulation and a manufacturing-ready package (Gerbers, BOM, Pick & Place, 174 parts).

Digital IIR Crossover Filter (Linkwitz-Riley LR4) on FPGA — *02/2026 – 06/2026*
SystemVerilog, Gowin GW1NR-9C (Tang Nano 9K). Designed a real-time 48 kHz/24-bit audio crossover in Q4.36 fixed point with 72-bit accumulators and an own-RTL UART at 921,600 bps. Verified bit-exact against a Python golden model; uses 12 % of registers and 8.7 % of LUTs.

Digital Oscilloscope with Real-Time Acquisition — *02/2026 – 06/2026*
SystemVerilog, Intel Cyclone IV (DE2-115). Team of 5. Designed the acquisition path and trigger logic: continuous capture at up to 60 MHz with zero sample loss through a circular RAM/FIFO buffer.

SoC Co-design: Nios II Processor with RTOS and Custom Peripherals — *02/2026 – 06/2026*
Quartus, C, Eclipse. Integrated Nios II soft-core in Platform Designer and custom peripherals in Verilog on the Avalon bus. Developed multitasking firmware in C using a Real-Time Operating System (RTOS).

Wireless Drip-Irrigation Controller (Bachelor's Thesis) — *04/2025 – 01/2026*
KiCad, ESP32-S2, ESPHome. Designed, fabricated and validated a 2-layer PCB: Wi-Fi module, 4 optocoupler-isolated relay outputs, 12 V→5 V buck converter (TPS54202) and a 4–20 mA level-sensor input with an RC low-pass filter (fc ≈ 5.9 Hz) added after measuring a noisy signal. ESPHome firmware with OTA updates and Home Assistant integration.

Work Experience

Industrial Electronic Maintenance Technician — *07/2024 – 08/2024*
Postres Lácteos Romar (Internship) — Catadau, Spain

- Performed preventive and corrective maintenance on automated machinery in a continuous production plant, diagnosing faults in sensors, actuators and electrical control systems.

Head Waiter — *01/2021 – Present*
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- Lead floor service at peak times (150+ diners) and train new waiting staff, while combining this job with a BSc and two MSc degrees for 5+ years.

Skills

RTL & Verification	Verilog, SystemVerilog (RTL and class-based testbenches), constrained-random stimulus, functional coverage, reference models, fault injection
Programming	C, C++, Python, Tcl, MATLAB/Simulink
EDA & FPGA	Intel Cyclone IV, Cyclone V (SoC), Tang Nano 9K (Gowin GW1NR-9C), Nios II soft-processor · Intel Quartus Prime, Gowin EDA, ModelSim, QuestaSim, Git
Interfaces & Buses	RTL: UART (own), Avalon, VGA · Board level: Ethernet (RMII PHY), CAN, USB 2.0 FS, SPI Flash
PCB & Embedded	Altium Designer (4-layer, controlled impedance), KiCad, STM32F746ZG board design, ESP32-S2 (ESPHome)
Languages	Spanish, Valenciano, English B1 (preparing B2 certification)